

MACHINE LEARNING-BASED EARTHQUAKE SEVERITY CLASSIFICATION

Progress Report

An Adapted Replication and Improvement of an Earthquake Classification Methodology

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Academic Year: 2026–2027

1. Abstract

This progress report presents the work completed so far on a machine learning-based earthquake severity classification project. The project is based on an adapted replication of an existing earthquake classification methodology. Two earthquake datasets have been prepared and combined, duplicate records have been identified and removed, and the data have been processed for machine learning.

Because the available datasets contain earthquake magnitudes from 6.5 to 9.1, the original paper's magnitude classes cannot be reproduced exactly. An adapted three-class severity classification has therefore been defined. Data preprocessing, feature engineering, chronological splitting, class-balancing methodology, and several machine learning algorithms have been prepared and explored. The model is still under development, so no accuracy or other final performance values are reported in this progress report.

2. Introduction

Earthquakes are natural hazards that can cause serious damage to people, buildings, infrastructure, and the environment. Machine learning can help analyze historical earthquake information and identify patterns related to earthquake severity.

The present project aims to develop a machine learning system for earthquake severity classification using historical earthquake data. The current stage focuses on dataset preparation, adaptation of the reference methodology, preprocessing, feature engineering, and development of the machine learning pipeline. Final model optimization and validation are still pending.

3. Problem Statement

The objective is to develop a machine learning-based system that can classify earthquake events according to severity. The available datasets contain missing values, duplicate events, different column structures, and an imbalanced distribution of earthquake magnitudes. In addition, some variables used in the reference paper are not available in the current datasets. Therefore, the original methodology needs to be adapted to the available data.

4. Objectives

- Prepare and clean the available earthquake datasets.
- Combine the datasets and remove duplicate earthquake events.
- Analyze seismic, geographic, and catalog-related attributes.
- Create an appropriate target variable using the available magnitude range.
- Perform data preprocessing and feature engineering.
- Use a chronological train-test methodology.
- Handle class imbalance using SMOTE within the training process.
- Develop and compare multiple machine learning algorithms.
- Use explainability methods to understand important features.
- Complete model optimization and final validation in the next stage.

5. Reference Methodology and Adaptation

The reference paper uses earthquake-related features and machine learning algorithms for earthquake classification. Its methodology includes magnitude-based classification, data preprocessing, class balancing, chronological evaluation, and multiple machine learning classifiers.

An exact replication is not possible with the present datasets because they contain only earthquakes with magnitude ≥ 6.5 and do not contain every feature used in the reference methodology. In particular, some variables reported in the paper, such as Close distance and RMS, are not available. Therefore, the present work is treated as an adapted replication rather than an exact reproduction.

6. Dataset Description

Two earthquake datasets are being used.

Dataset 1 contains 1,000 records and 19 columns. Dataset 2 contains 782 records and 19 columns. The datasets include information such as magnitude, date/time, CDI, MMI, alert, tsunami, significance, network, station count, distance measures, magnitude type, depth, latitude, longitude, location, continent, and country.

After combining the datasets, 1,782 records were available before duplicate removal. Duplicate checking using date/time, magnitude, latitude, and longitude identified 784 duplicate records. After removing these duplicates, 998 unique earthquake events remained for the modeling workflow.

The available magnitude distribution is:

- 6.5–<7.0: 695 events
- 7.0–<8.0: 279 events
- ≥ 8.0 : 24 events

This distribution shows that the highest-magnitude class is relatively small and therefore class imbalance is an important consideration.

7. Adapted Target Variable

The original paper's four magnitude-based categories cannot be directly applied because the current datasets contain only earthquakes with magnitude ≥ 6.5 . An adapted three-class target has therefore been created:

Class 1: 6.5–<7.0

Class 2: 7.0–<8.0

Class 3: ≥ 8.0

Magnitude itself is excluded from the predictor variables because it directly defines the target class. This helps prevent direct target leakage.

8. Data Preprocessing

- Loaded the available earthquake datasets.
- Checked and standardized the dataset structure.
- Parsed date/time information.
- Converted numerical fields to appropriate numeric types.
- Checked missing values and prepared the data for imputation.
- Identified and removed duplicate earthquake events.
- Sorted records chronologically.
- Created the adapted three-class target variable.

9. Feature Engineering

The current feature-engineering stage uses seismic, geographic, catalog, and time-derived information. Candidate features include latitude, longitude, depth, station count (nst), gap, dmin, cdi, mmi, tsunami, and sig.

Additional engineered variables include absolute latitude and longitude, logarithmic transformations of selected skewed variables, square-root depth, year, month, day of year, hour, cyclical time features, and selected interaction features.

Some variables such as CDI, MMI, and SIG are catalog/post-event variables. Therefore, the current project should be described as catalog-based earthquake severity classification rather than a strict real-time early-warning system.

10. Experimental Methodology

A chronological train-test strategy has been prepared so that earlier earthquake records are used for training and later records are used for testing. This is more appropriate for a time-dependent earthquake dataset than simply performing a random split.

SMOTE has been incorporated into the training workflow to address class imbalance. It is applied to training data rather than the held-out test data to reduce the risk of information leakage.

The model-development pipeline also includes preprocessing, feature transformation, model training, hyperparameter optimization, evaluation, and explainability.

11. Machine Learning Algorithms

- Random Forest
- Support Vector Machine (SVM)
- Multi-Layer Perceptron (MLP)
- XGBoost
- LightGBM

12. Current Development Status

Component	Current Status
Dataset loading and preparation	Completed / implemented
Dataset combination	Completed / implemented
Duplicate identification and removal	Completed / implemented
Target-class adaptation	Completed / implemented
Preprocessing pipeline	Implemented; further checking required
Feature engineering	Implemented
Chronological train-test strategy	Implemented
SMOTE class balancing	Implemented
Machine learning models	Implemented / under testing
Hyperparameter optimization	In progress
Final model selection	Pending
Final validation	Pending
Final performance reporting	Pending
SHAP/explainability	Explored; final analysis pending

13. Explainability Work

SHAP-based feature analysis has been explored to understand the contribution of input variables to model predictions. The explainability stage will be finalized after the model pipeline is stabilized and the final model has been selected.

14. Current Limitations

- The available datasets contain only magnitude ≥ 6.5 events, preventing exact reproduction of the paper's original classes.
- Some variables used in the reference paper are not available.
- The dataset is relatively small, especially for the ≥ 8.0 class.
- The model pipeline is still being debugged and optimized.
- Final validation has not yet been completed.
- Some available features are post-event/catalog variables and are not appropriate for strict real-time early warning.

15. Work Remaining

1. Debug and stabilize the complete training pipeline.
2. Verify missing-value handling and preprocessing.
3. Complete hyperparameter tuning.
4. Perform consistent final model comparison.
5. Generate final confusion matrices and classification reports.
6. Validate the selected model on the held-out chronological test set.
7. Finalize SHAP/explainability analysis.

8. Save the final trained model and preprocessing pipeline.
9. Prepare final figures and tables.
10. Update this progress report into the final project/research report.

16. Conclusion –

Significant progress has been made in the earthquake severity classification project. The datasets have been prepared, combined, cleaned, and adapted for the available magnitude range. A three-class target has been created, preprocessing and feature-engineering pipelines have been developed, and multiple machine learning algorithms have been incorporated into the experimental workflow.

However, the model is not yet complete. Final optimization, validation, model selection, explainability, and reproducibility checks remain to be completed. For this reason, no accuracy, F1-score, MCC, ROC-AUC, or other final performance values are included in this progress report.

17. Proposed Final Workflow

Dataset Collection → Data Cleaning → Duplicate Removal → Exploratory Data Analysis → Target Creation → Feature Engineering → Chronological Split → Training Data Balancing → Model Training → Hyperparameter Optimization → Final Evaluation → Explainability → Final Validation → Final Model → Final Report

18. References

1. Reference earthquake classification paper provided for this project.
2. Scikit-learn documentation and machine learning methodology references.
3. Documentation for XGBoost, LightGBM, and imbalanced-learn/SMOTE used in the experimental pipeline.

STATUS: WORK IN PROGRESS.....